

Architecture Document

I. Application Architecture

1.1 Microservices Architecture: The application will be divided into multiple microservices, each responsible for a specific functionality. The microservices will be designed to be loosely coupled, allowing for scalability and flexibility.

- **Market Data Service:** Fetches real-time and historical financial data from external APIs like Yahoo Finance.
- **Strategy Engine Service:** Implements and runs algorithmic trading strategies using the Adam optimization algorithm.
- **Execution Service:** Automatically places buy/sell orders based on signals generated by the strategy engine.
- **Risk Management Service:** Monitors risk exposure, including metrics like drawdown and Sharpe ratio, adjusting strategies dynamically.
- **Monitoring & Reporting Service:** Provides real-time analytics and performance tracking of trading activities.

1.2 Event-Driven Architecture: The system follows an event-driven design to trigger actions based on market events.

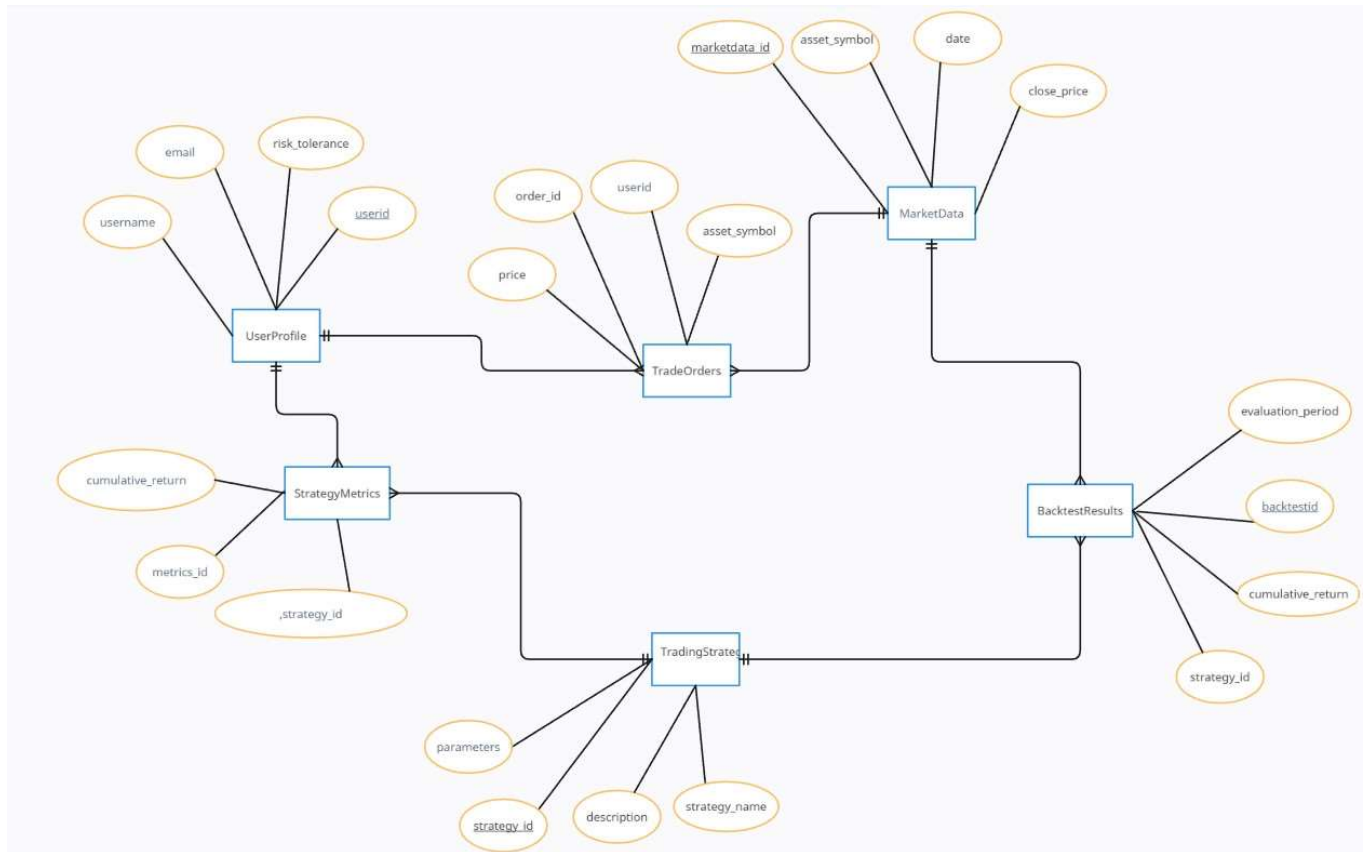
- **Market Data Events:** Triggered when new data is available, processed by the strategy engine to make trading decisions.
- **Trade Execution Events:** Sent when a trade signal is generated, causing the Execution Service to place orders.
- **Risk Alert Events:** Triggered if predefined risk thresholds (e.g., high drawdown) are exceeded, initiating an adjustment in strategy.

1.3 Serverless Components: Certain components, such as backtesting and report generation, run on serverless platforms like AWS Lambda. These tasks are executed on-demand, optimizing resource utilization:

- **Backtesting Functions:** Runs simulations on historical data using the Adam algorithm to test trading strategies.
- **Analytics Reporting:** Periodically generates performance reports, leveraging serverless functions for quick, scalable execution.

II. Database Architecture

2.1 ER Diagram



2.2 Schema Design

The schema design will consist of the following tables:

- stocks:

id	symbol	name	exchange
(primarykey)			

- historical_prices:

id	stock_id	date	price
(primary key)	(foreign key referencing stocks.id)		

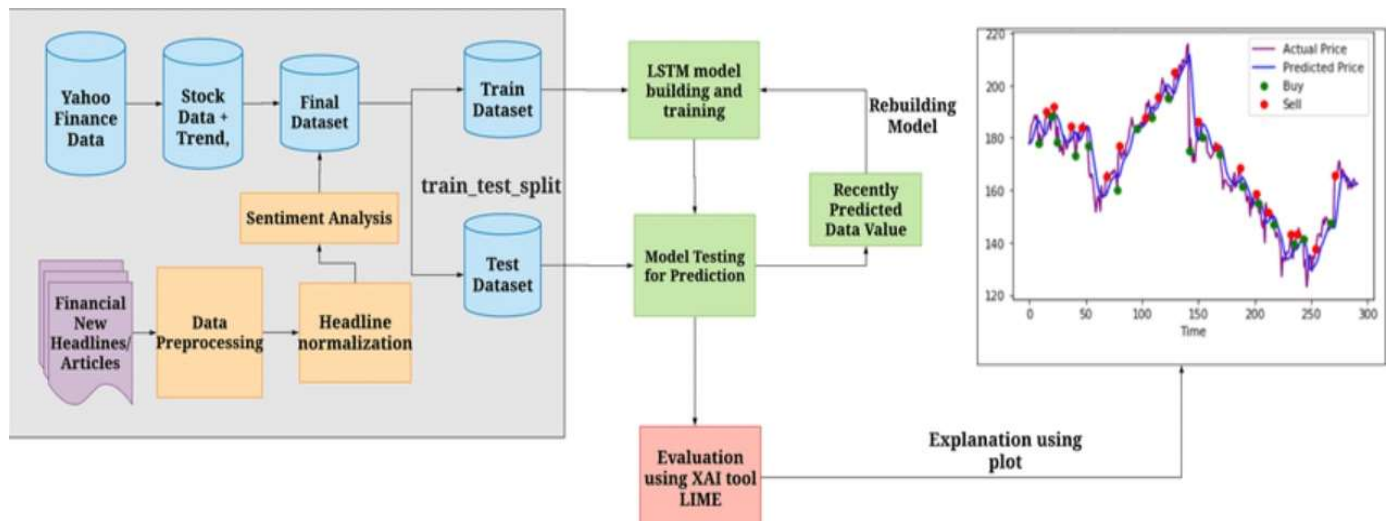
- processed_data:

id	stock_id	returns	date
(primary key)	(foreign key referencing stocks.id)		

- model_predictions:

id	stock_id	date	prediction
(primary key)	(foreign key referencing stocks.id)		

2.3: System Architecture Diagram



Data Sources:

- **Yahoo Finance Data:** Historical stock prices and trends are sourced from Yahoo Finance.
- **Stock Data + Trend:** This includes numerical stock prices and the overall trend direction over time, essential for time series forecasting.
- **Financial News Headlines Articles:** News articles that may influence stock price movements, providing sentiment data.

Data Preprocessing:

- **Data Preprocessing:** Involves cleaning and preparing both numerical stock data and textual financial headlines.
- **Headline Normalization:** Standardizing or normalizing news headlines to reduce noise and extract relevant sentiment-related information.

Sentiment Analysis:

- **Sentiment Analysis:** This process extracts sentiment (positive, negative, neutral) from the financial headlines to understand how news may affect stock trends. This data is then integrated into the final dataset.

Modeling and Prediction:

- **Train Dataset / Test Dataset:** The dataset is split into training and testing sets, with the training set used to train the LSTM model and the test set used for prediction evaluation.
- **LSTM Model Building and Training:** The Long Short-Term Memory (LSTM) model is trained on the stock data, combined with sentiment data, to predict future stock prices.
- **Model Testing for Prediction:** The trained model is tested with unseen data (test dataset) to evaluate its predictive performance.
- **Rebuilding Model:** If necessary, the model is refined and retrained based on the results.

Evaluation and Results:

- **Evaluation using XAI Tool LIME:** The predictions are explained using the Local Interpretable Model-agnostic Explanations (LIME) tool, which helps interpret the results of complex models like LSTM by simplifying the predictions.
- **Explanation using Plot:** The graph in the image compares the actual stock price with the predicted stock price. Buy and sell signals are also visualized with red and green markers.

III. Data Exchange Contract

3.1 Frequency of Data Exchanges

- **Market Data:** Fetched every minute or in real-time.
- **Trade Execution:** Immediate when signals are generated.
- **Risk Monitoring:** Continuously, adjusting based on predefined risk thresholds.

3.2 Data Sets

- **Historical Prices:** symbol, date, price
- **Processed Data:** stock_id, date, returns
- **Model Predictions:** stock_id, date, prediction

3.3 Mode of Exchanges

- **APIs:** Real-time market data is pulled through REST APIs from financial data providers.
- **File Exchange:** Historical market data may be ingested through CSV files for backtesting.
- **Message Queues:** Events like trade signals and risk alerts are communicated through message queues like Kafka for real-time processing.